# Results: Language, Topics, and Embeddings

## RQ3: Topical and Linguistic Structure I

Text analysis operates over titles, abstracts, and (when available) full text. A TF-IDF representation over a 137-feature vocabulary feeds non-negative matrix factorization, which extracts 5 latent topics cross-cutting the subfield taxonomy. The top vocabulary terms are: atmospheric, uncertainty, spectra, temperature, abundance, analysis, assumptions, opacity, retrievals, sources, stellar, coverage, evaluated, observed, across, composition, models, provide, competing, retrieval.

**Table 3. NMF topics extracted from the corpus.**

# RQ3: Topical and Linguistic Structure I

| Topic | Top terms |
|---|---|
| 0 | atmospheric, cloud, retrieved, species, changes, structure, models, explore |
| 1 | prior, methane, depend, spectral, dioxide, simulated, carbon, constrained |
| 2 | contributions, separate, resolution, planetary, high, systematics, modelled, instrument |
| 3 | population, reporting, studies, preprocessing, require, harmonized, separates, specific |
| 4 | model, intervals, misspecification, noise, quantify, effect, record, dependencies |

## RQ3: Topical and Linguistic Structure I

The topic labels are computed from the retained corpus and are not hand-assigned. Their interpretation should be checked against the generated topic-term weights and the configured subfield taxonomy; fixture-derived topics describe synthetic text patterns, not the scientific field.

# RQ3: Topical and Linguistic Structure I

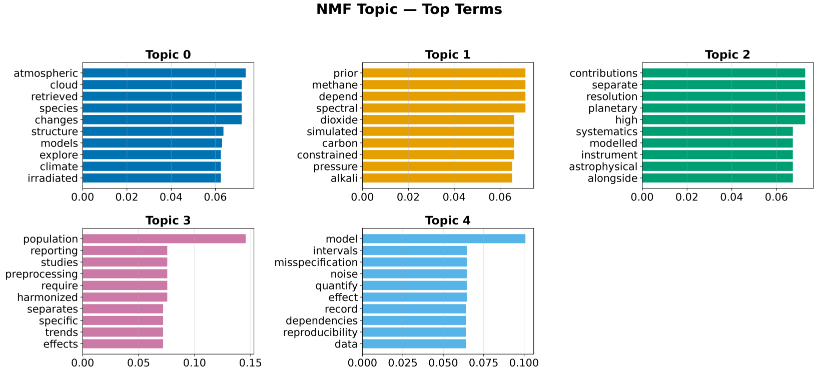
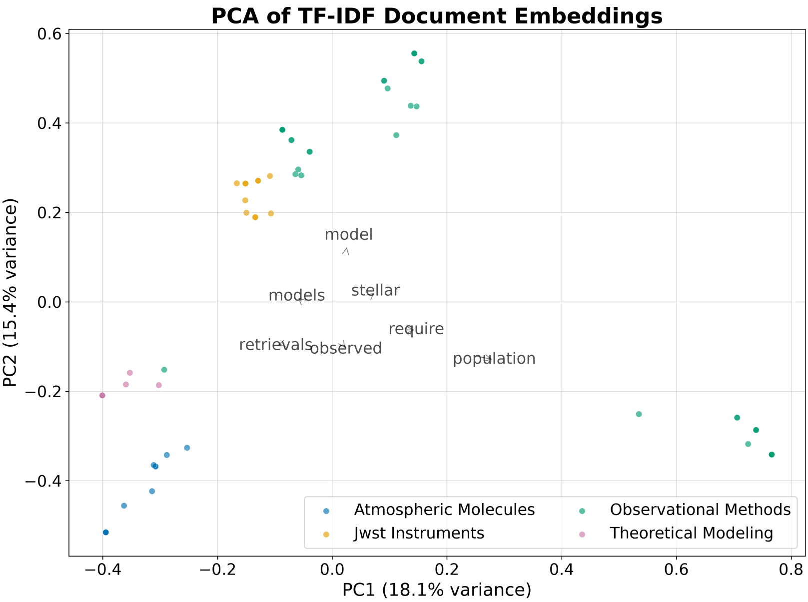



Figure 1: Topic-term bar charts for Exoplanet Atmospheres. Each panel shows the top weighted terms for one of 5 NMF topics, with bar length proportional to the topic-term weight in the $\mathbf{H}$ matrix.

# Document Embeddings I

Offline deterministic embeddings (TF-IDF followed by truncated SVD) place every document in a shared 50-dimensional vector space. Embedding the same text twice yields identical vectors, so the derived similarity matrix, nearest-neighbour lists, clusters, and two-dimensional projection are all reproducible.

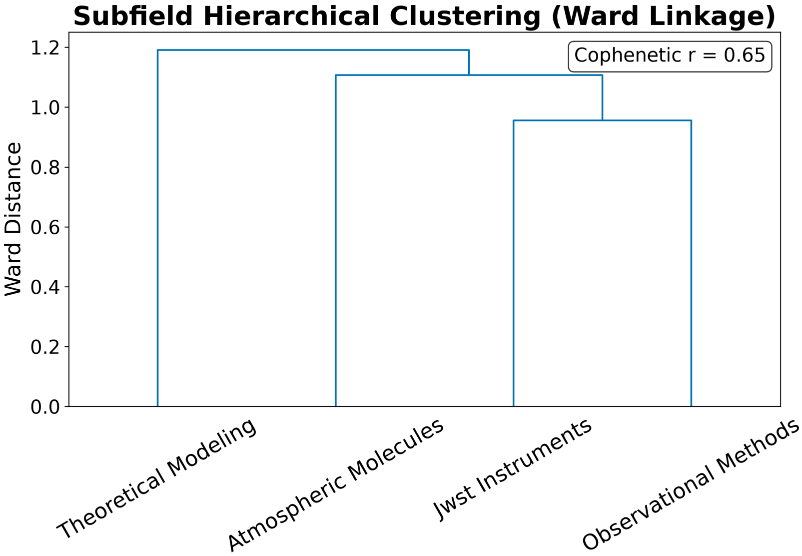
# Document Embeddings I



# Document Embeddings II

Figure 2: PCA projection of document embeddings for Exoplanet Atmospheres. Each point represents one document projected onto the first two principal components of the TF-IDF/SVD embedding. Colours indicate subfield assignment, showing how the topical geography relates to the keyword taxonomy.

# Document Embeddings III



# Document Embeddings IV

Figure 3: Hierarchical clustering dendrogram of document embeddings. The tree shows the similarity structure of the corpus: documents that join low in the tree are semantically similar, while high-level splits separate the major topical clusters.

## Term Analysis I

The TF-IDF term heatmap reveals which terms discriminate between subfields: terms with high between-subfield variance (rather than high global mean) are selected for display.

# Term Analysis I

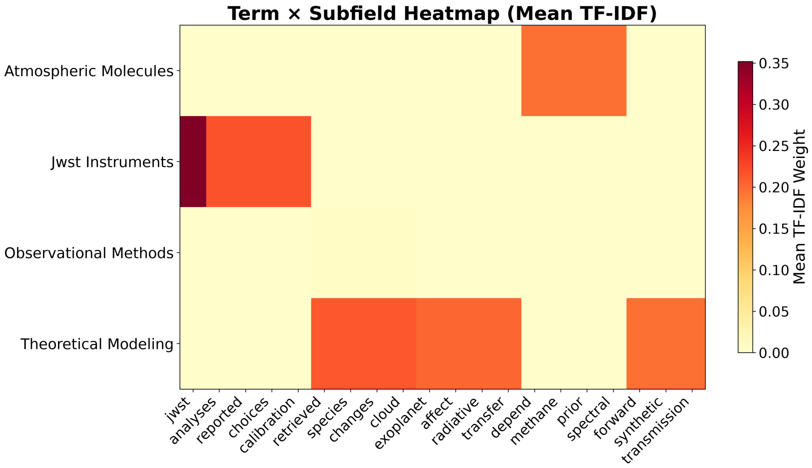



Figure 4: Term heatmap for Exoplanet Atmospheres. Each cell shows the mean TF-IDF weight of a term within a subfield. Terms are selected by between-subfield variance to highlight discriminative vocabulary rather than globally frequent terms.

# Term Analysis I

Named entity extraction over the 80 abstracts identified 1 unique entities. The most frequent entities reflect the retained corpus and the configured extraction rules; source coverage and fixture status determine how they may be interpreted.

**Table 4. Top named entities in abstracts.**

# Named Entity Analysis I

| Entity | Frequency |
|---|---|
| JWST | 17 |

# Named Entity Analysis II

| Entity | Frequency |
|---|---|
| JWST | 17 |

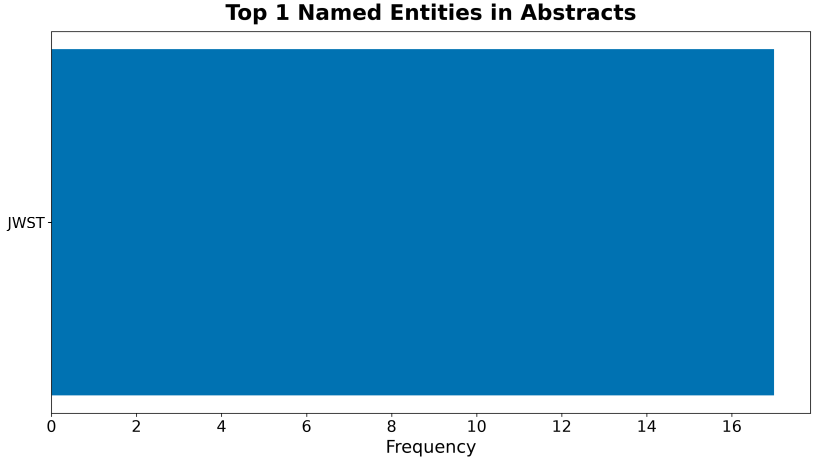



Figure 5: Top named entities for Exoplanet Atmospheres. The horizontal bar chart shows the 20 most frequently

# Named Entity Analysis I

# Named Entity Analysis II

**Table 5. Top keyphrases by TF-IDF score.**

# Named Entity Analysis I

| Keyphrase | Score |
|---|---|
| jwst | 0.0714 |
| population | 0.0556 |
| analyses | 0.0435 |
| calibration | 0.0435 |
| calibration choices | 0.0435 |
| choices | 0.0435 |
| combined | 0.0435 |
| jwst analyses | 0.0435 |
| models | 0.0435 |
| models calibration | 0.0435 |
| models calibration choices | 0.0435 |
| model | 0.0400 |
| atmospheric | 0.0392 |
| abundance | 0.0385 |
| assumptions | 0.0385 |

The TF-IDF/SVD embeddings place every document in a 50-dimensional vector space. K-means clustering with $k = 5$ clusters partitions the corpus into topically coherent groups. The top similar document pairs, ranked by cosine similarity, reveal the most closely related works in the corpus.

**Table 6. Top 10 most similar document pairs.**

# Embedding Similarity and Clustering I

| Paper A | Paper B | Similarity |
|---|---|---|
| doi:10.5555/exoplanet atmosphe | doi:10.5555/exoplanet atmosphe | 1.0000 |
| doi:10.5555/exoplanet atmosphe | doi:10.5555/exoplanet atmosphe | 1.0000 |
| doi:10.5555/exoplanet atmosphe | doi:10.5555/exoplanet atmosphe | 1.0000 |
| doi:10.5555/exoplanet atmosphe | doi:10.5555/exoplanet atmosphe | 1.0000 |
| doi:10.5555/exoplanet atmosphe | doi:10.5555/exoplanet atmosphe | 1.0000 |
| doi:10.5555/exoplanet atmosphe | doi:10.5555/exoplanet atmosphe | 1.0000 |
| doi:10.5555/exoplanet atmosphe | doi:10.5555/exoplanet atmosphe | 1.0000 |
| doi:10.5555/exoplanet atmosphe | doi:10.5555/exoplanet atmosphe | 1.0000 |
| doi:10.5555/exoplanet atmosphe | doi:10.5555/exoplanet atmosphe | 1.0000 |
| doi:10.5555/exoplanet atmosphe | doi:10.5555/exoplanet atmosphe | 1.0000 |

# Embedding Similarity and Clustering II

| Paper A | Paper B | Similarity |
|---|---|---|
| doi:10.5555/exoplanet atmosphe | doi:10.5555/exoplanet atmosphe | 1.0000 |
| doi:10.5555/exoplanet atmosphe | doi:10.5555/exoplanet atmosphe | 1.0000 |
| doi:10.5555/exoplanet atmosphe | doi:10.5555/exoplanet atmosphe | 1.0000 |
| doi:10.5555/exoplanet atmosphe | doi:10.5555/exoplanet atmosphe | 1.0000 |
| doi:10.5555/exoplanet atmosphe | doi:10.5555/exoplanet atmosphe | 1.0000 |
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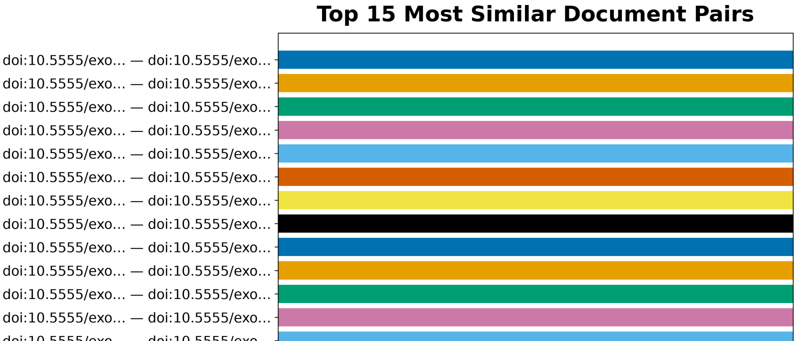
| doi:10.5555/exoplanet atmosphe | doi:10.5555/exoplanet atmosphe | 1.0000 |
| doi:10.5555/exoplanet atmosphe | doi:10.5555/exoplanet atmosphe | 1.0000 |



## Embedding Similarity and Clustering I

These embeddings support semantic retrieval over the corpus and the visual map of the literature's topical geography.